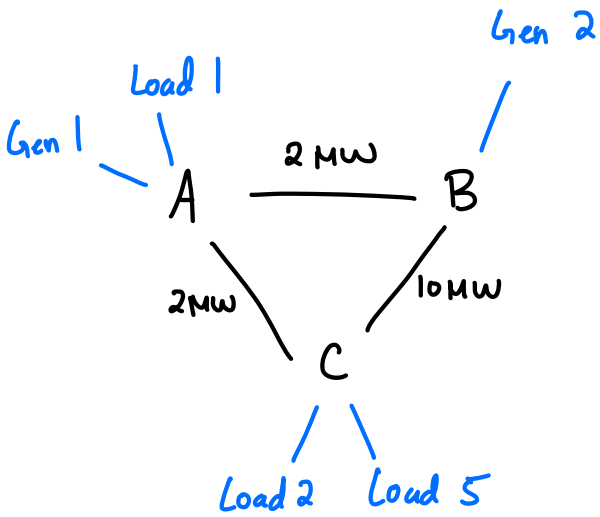
1) DA Clearance



Note that at each Settlement Point, buyers and Sellers can transact an unlimited amount of power.

- At node A, 1 MW is demanded
- At node B, 0 MW is demanded
- At node C, 3 MW is demanded.

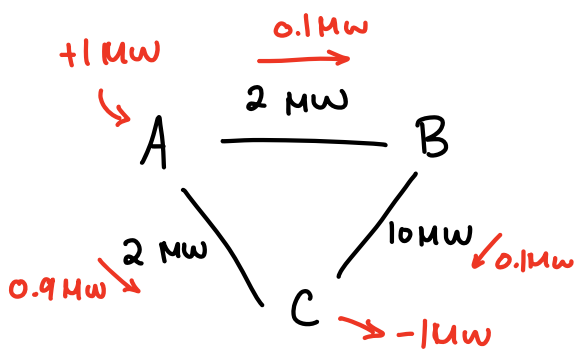
Note that when Gen 1 produces 1 MW, it cannot Control which line the power flows through. Instead, the 1 MW generated by gen 1 flows through each line according to their shift factors. (determined by the electrical Properties of the transmission lines (voltage, distance, etc).

The Calculation of these Shift factors are done by Powerworld.

For the Sake of example, Suppose Shift factors are: (with 1 MW injected into A and withdrawn from C).

A–B	B–C	A–C
0.1	0.1	0.9

So, 1 MW injected in A and withdrawn in C flows as follows



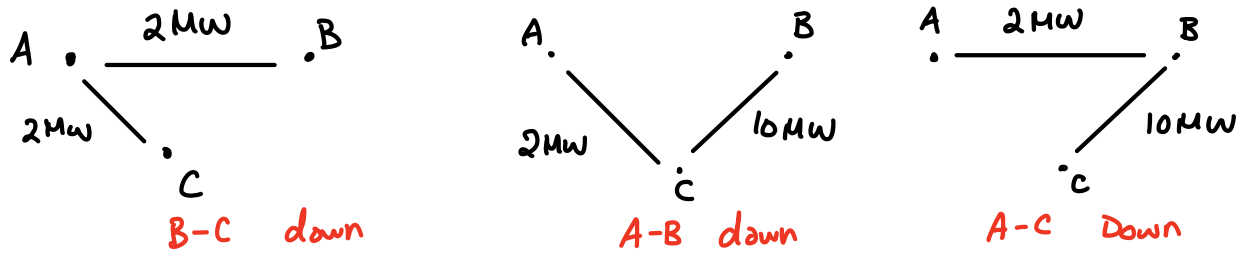
We calculate Shift factors for all other Combinations of injection/withrawl points to calculate what the total flow will be.

Note the ISO balances the grid by settling the auction, such that willing buyers buy the most electricity at the best prices. Then the grid provides a feasible generation / consumption schedule to cleared participants.

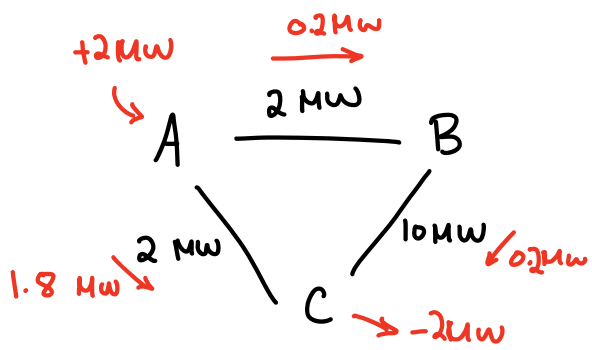
Consider the following clearing logic, which is easier to understand

- Load 1 is willing to pay up to \$6.50, but max offer price is only \$5, so load 1 will pay only \$5.00/MW.
- Load 2 willing to pay \$7.50. It looks like it can buy cheaper power at \$5 from gen 1, but is being outcompeted by load 3. However, it looks like there is 0.5 MW load 2 can still buy from Gen 1. Can it really?

Here we have to consider shift factors and the N-1 contingency check. That is, the final allocation has to be feasible even if every possible 1-line outage occurs. That is, our solution has to be feasible in:



Note we are constrained by the A-B down case, with Gen 1 generating 3 MW. 1 MW is consumed by load 1 directly connected, and the remaining 2 MW flows as follows:



Note that even though 0.2 MW can still flow into A-C (so 0.2/0.9 = 0.22 MW more can be generated, if any line goes down, then the system will overload, potentially resulting in catastrophic failure.

Auction Clearance

Therefore, the Clearance at node C Sorts bidders by price, and allocates the 2 MW to 0.5 MW (from Gen 3) to customers.

Load 3	1.5 MW	\$10.00/MW	[fully Cleared]
Load 2	2 MW	\$7.50/MW	[partially Cleared for remaining 1 MW]

And importantly, the auction clears at \$7.50/MW, despite the fact that 2/2.5 MW came from cheaper Gen 1.

Load 3 was willing to pay for \$10.00 too, but ended up not being the minimum Cleared bid, so it only needs to pay the final clearing price of \$7.50/MW too, just as a result of the Dutch auction mechanism.